

MKGF: A multi-modal knowledge graph based RAG framework to enhance LVLMs for Medical visual question answering

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ABSTRACT

Medical visual question answering (MedVQA) is a challenging task that requires models to understand medical images and return accurate responses for the given questions. Most recent methods focus on transferring general-domain large vision–language models (LVLMs) to the medical domain by constructing medical instruction datasets and in-context learning. However, the performance of these methods are limited due to the hallucination issue of LVLMs. In addition, fine-tuning the abundant parameters of LVLMs on medical instruction datasets is high time and economic cost. Hence, we propose a MKGF framework that leverages a multi-modal medical knowledge graph (MMKG) to relieve the hallucination issue without fine-tuning the abundant parameters of LVLMs. Firstly, we employ a pre-trained text retriever to build question–knowledge relations on training set. Secondly, we train a multi-modal retriever with these relations. Finally, we use it to retrieve question-relevant knowledge and enhance the performance of LVLMs on the test set. To evaluate the effectiveness of MKGF, we conduct extensive experiments on two public datasets Slake and VQA-RAD. Our method improves the pre-trained SOTA LVLMs by 10.15% and 9.32%, respectively. The source codes are available at <https://github.com/ehnal/MKGF>.

1. Introduction

Medical Visual Question Answering (MedVQA) aims to build a model that can deeply understand medical images and questions, and provide accurate responses. For example, given a medical image of “pneumonia” and a question “Is there the abnormality in this image?”, MedVQA requires models to output the answer “Yes”. Due to its ability to analyze medical images and answer medical questions, MedVQA is an important task that helps provide valuable medical suggestions to patients and alleviates the workload of doctors [1].

Limits of previous methods. According to modeling approaches, existing MedVQA solutions can be mainly divided into two categories: (1) discriminative and (2) generative methods [2].

The first category treats MedVQA as a classification problem, where models are required to predict answers from a pre-defined set. Works along this line are dedicated to improving cross-modal interaction with attention mechanisms [3–5] or relieving data scarcity by introducing external medical image–caption datasets [6–9]. Although these methods have good performance on a specific dataset, they are limited to making closed-set predictions [10]. This limitation makes it challenging for them to adapt to diverse questions and practical

application scenarios, where predefined candidate answers are often unavailable [2,11]. To solve these problems, the second category of methods has been proposed, which models MedVQA as a generation task. With the advancement of Large Vision–Language Models (LVLMs), a mainstream idea is to transfer LVLMs into the medical domain (e.g., LLava-med [2]). Works along this line mainly rely on existing medical image–caption datasets (e.g., ROCO [12]) to construct instruction training examples [2,13–15] and design in-context learning or thought chain prompts for the LVLm [16–19]. However, these methods face two main challenges: (1) Similar to their performance in the general domain, LVLMs in the medical field also suffer from the hallucination problem. They may generate outputs that are inconsistent with factual information due to a lack of question-related knowledge. For example, as shown in Fig. 1(a), LLava-med returns an incorrect answer because of lacking certain knowledge related to *pneumonia*. (2) It is difficult to update the medical knowledge memorized by LVLMs. Finetuning the abundant parameters of LVLMs on new medical datasets or knowledge bases requires high time and economic cost.

Our idea and contribution. We argue that knowledge-based Retrieval Augment Generation (RAG) is a solution for the above two

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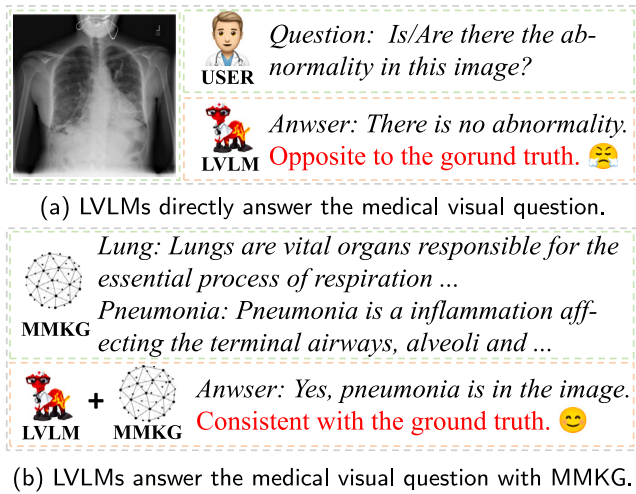


Fig. 1. Examples of how the retrieved knowledge benefits LVLMs for MedVQA. The LVLm in images is Llava-med.

issues. Firstly, RAG retrieves question-related knowledge from external medical knowledge bases as supplementary input for LVLMs, which helps mitigate hallucination problems. For example, as shown in Fig. 1, Llava-med revises its previous incorrect answer with the retrieved entity knowledge (e.g., pneumonia) from a medical knowledge graph. Secondly, RAG is a cost-effective method that can provide new knowledge to LVLm without finetuning its large number of parameters.

A straightforward approach is to transfer a general domain knowledge-base RAG framework (e.g., Wiki-LLava [20]) into the MedVQA task. However, the performance of this approach is limited for the following issues. (1) Retrieving question-relevant knowledge in MedVQA is more challenge than in the general domain. Firstly, general VQA mainly focuses on single entity knowledge, while MedVQA often involves knowledge about multiple related entities. For example, as shown in Fig. 1(a), the input medical image includes various organ or disease entities (i.e., lung and pneumonia). Secondly, images of general entities (i.e. cats and dogs) exhibit clear distinctions in color and shape. In contrast, medical images (e.g., X-rays) are typically black-and-white, and the shapes of organs may lack clarity. (2) Training a knowledge retriever for MedVQA is harder than in the general domain. General knowledge-based VQA datasets (e.g., Encyclopedic-VQA [21]) provide question–entity matching relationships for training the knowledge retriever. However, the MedVQA datasets (e.g., Slake [22]) do not provide this relationships.

Hence, in this paper, we construct a multi-modal medical knowledge graph (MMKG) and propose an MMKG-based RAG framework (MKGF) to address the above issues. The core idea is that we first use an LVLm and a pre-trained retriever to obtain question–entity matching relationships on the training set. These relationships are then employed to train a multi-modal retriever, which searches entity knowledge from the MMKG to supplement the LVLm input during testing. Specifically, we first employ an LVLm to generate the rationales for the medical visual question–answer pairs in the training set. Then we compute the similarity between these rationales and entities using a pre-trained retriever (e.g., BGE [23]). Since medical images often involve multiple related entities, we design a sub-graph matching algorithm to obtain question–entity mappings, which considers the entity–rationale similarity of both the entity itself and its neighbors. After obtaining these mappings, we employ them and an MMKG to train a multi-modal retriever with two loss functions. The first contrastive loss assists the retriever in distinguishing the images of different entities in MMKG. The second KL divergence loss employs the mappings to help the model understand the entity–question matching relationships. Finally, given

a medical visual question in the test set, we employ the trained multi-modal retriever to get the relevant entity knowledge from the MMKG. The retrieved knowledge is regarded as the supplementary input of the LVLm to relieve the hallucination issue. Our main contributions are summarized as follows:

- We propose a novel MedVQA framework named MKGF. To the best of our knowledge, this is the first work to combine MMKG and LVLm for the MedVQA task.
- We construct a new multi-modal medical knowledge graph, which contains 20,545 images of 409 organ or disease entities. We believe it a valuable resource for future works in MedVQA.
- We conduct extensive experiments on two public MedVQA datasets Slake and VQA-RAD. The proposed MKGF improves the SOTA medical LVLm by 10.15% and 9.32% on these datasets respectively.

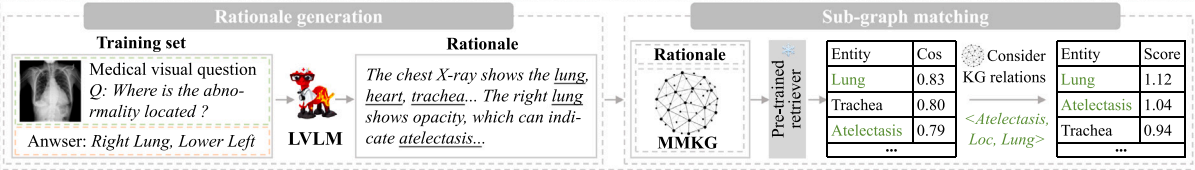
2. Related work

MedVQA methods. Existing works mainly model MedVQA as a classification task or a generation task. The classification methods focus on two main issues: multi-modal information processing and data scarcity. To solve the first issue, some works design an attention layer [3,5] or transformer structure [4] for multi-modal feature interaction. To solve the second issue, some studies introduce external medical image–caption datasets to supplement training samples with relevant multi-modal information [6] or to perform large-scale pre-training [7–9]. Recently, inspired by the success of graph networks in processing visual information [24,25] and assisting LLMs [26] for general-domain multi-modal tasks, some works researches on transferring graph networks into medical domain for radiology summarization [27,28] and MedVQA [22,29]. However, these MedVQA methods only make close-end predictions, which are limited to practical application scenarios. Hence, some works trend to build MedVQA as a generative task to solve the above issue [11,30,31]. With the development of LVLMs, recent works aim to transfer general-domain LVLMs into medical domains through two main ways. The first category focuses on constructing high-quality medical instruction datasets to fine-tune LVLMs [2,13–15]. Some works additionally introduce chain-of-thought information [17] and relative radiology reports [32] for the question–answering pairs to train a stronger medical LVLm. The second category provides LVLMs with few-shot cases [18] or thought chain design examples [16,19] to enhance its ability on MedVQA. However, fine-tuning LVLMs involves high costs, and the performance of LVLMs is limited on MedVQA due to the hallucination problem.

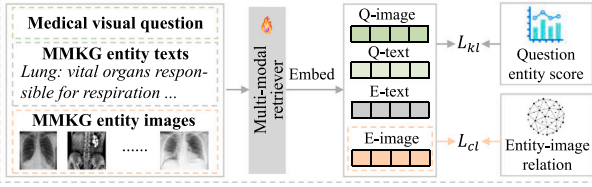
Knowledge-based multi-modal RAG. The main challenge of knowledge-based RAG is to bridge the visual question with named entities in knowledge base. Existing methods can be mainly classified into three categories: The first category focuses on leveraging the region features of the question image, which are extracted with a slide window [33] or object detection models [34,35]. The second category is dedicated to processing the entity text information, ranging from the titles [36] to chunks [20]. The third category emphasizes to calculate the multi-modal similarity of entities and visual questions from multiple perspectives, including cross-modal and mono-modal computing [37,38]. These methods are primarily designed to address visual questions involving a single entity. However, MedVQA involves knowledge related to multiple entities, and its datasets lack question–entity pair annotations for training retrievers [22].

MedVQA data resources. Existing external data resources for MedVQA can be classified into two main types (1) medical image–caption datasets and (2) medical knowledge graphs (MKGs). The first type provides medical images and their corresponding textual descriptions to help the model align medical multi-modal information [12,13,39]. However, they do not provide descriptions of medical entities, which are necessary knowledge for answering medical questions [40]. Hence,

Step1: Question-entity similarity computing



Step2: Multi-modal knowledge retriever training



Step3: Retrieval augmented generation

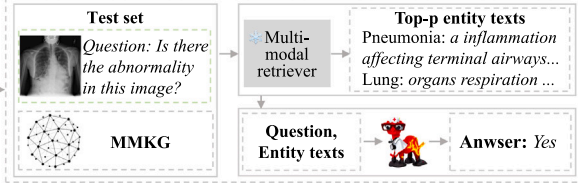


Fig. 2. The framework of MKGF which mainly consists of three parts: (1) question–entity similarity computing, (2) multi-modal knowledge retriever training and (3) retrieval augmented generation.

recent works build medical knowledge graph (MKG) to help models better understand medical entities and their relations [22,29]. However, MKGs lack visual information, which limits the model’s ability to retrieve and utilize entity knowledge for MedVQA.

3. Overview

In this section, we first formulate the MedVQA task in this paper and then outline our proposed framework MKGF.

3.1. Problem formulation

We first introduce some important concepts in this paper and then formulate the MedVQA task.

Multi-modal medical knowledge graph (MMKG). An MMKG $G = \{E, R_e, T, I\}$ consists of numerous medical entities $E = \{e_1, e_2, \dots, e_n\}$, relations between entities R_e , entities’ textual descriptions $T = \{T_1, T_2, \dots, T_n\}$ and images $I = \{I_1, I_2, \dots, I_n\}$. Specifically, for each entity $e_i \in E$, it has the corresponding textual description T_i , image I_i and a neighboring entity set E_i , where all the entity $e_j \in E_i$ has a relation $(e_i, r_{ij}, e_j) \in R_e$ with e_i . For example, given the relation (Alveolar Proteinosis, location, Lung), “Alveolar Proteinosis” and “Lung” are neighboring entities.

MedVQA task formulation. Given a pair of medical image and question (Q_i, P_i) , MedVQA task requires the model to return a correct answer A_i of the question Q_i based on the image P_i . For example, given a medical image of “pneumonia” and a question “Is the abnormality in this image?”, the expected output of the model is “Yes”. In this paper, we expect to introduce the entities’ textual descriptions T from an MMKG as external knowledge to assist the LVLM in answering medical questions. Hence, MedVQA can be formulated as $A_i = LVLM(Q_i, P_i, T)$.

3.2. Framework

As depicted in Fig. 2, MKGF mainly consists of three modules: (1) question–knowledge similarity computing, (2) multi-modal knowledge retriever training and (3) retrieval augmented generation.

Question–knowledge similarity computing. Given an MMKG, a visual question and its answer in the training set, we first employ an LVLM to generate the rationale for the given question–answer pair. Then we design a sub-graph algorithm and introduce a pre-trained retriever (e.g., BGE [23]) to compute the question–entity similarity scores based on the generated rationale and entity descriptions.

Multi-modal knowledge retriever training. After obtaining the question–entity similarity scores, we employ them and entity images in the MMKG to train a multi-modal retriever (e.g., BiomedCLIP [8]).

The training objective consists of a KL divergence loss function and a contrastive learning loss function, which help the retriever build the entity–question matching and distinguishing images of different entities, respectively.

Retrieval augmented generation. Given a visual question in the test set, we first use the trained multi-modal retriever to search for question-relevant entity descriptions in the MMKG. We then feed the visual question and these descriptions into an LVLM to obtain the final answer.

4. Methodology

In this section, we describe the details of the three main parts in our MKGF framework.

4.1. Question–entity similarity computing

Given an MMKG and the triple consisting of the medical image, question, and corresponding answer in the training set, this section aims to obtain the similarity scores between the given question and entities in the MMKG, which serve as supervision signals for the subsequent training of the multi-modal knowledge retriever. To this end, we first employ an LVLM to generate a rationale for the question based on the corresponding answer. Then we introduce a pre-trained textual knowledge retriever to compute the cosine similarity between the rationale and the entity descriptions. Finally, we design a sub-graph matching algorithm to obtain the question–entity similarity scores, which considers the relevance of both the entity itself and its neighboring entities to the rationale. The details are as follows.

Rationale generating. Given a triple of the question, image and corresponding answer (Q_i, P_i, A_i) in the training set D_{train} , we first concatenate them with the instruction prompt “Please generate a rationale based on the given question and answer”. Then we feed the whole prompt into an LVLM and obtain the rationale of (Q_i, P_i, A_i) . The process is formulated as:

$$R = \{R_i | R_i = LVLM(IP, Q_i, P_i, A_i), Q_i \in D_{train}\} \quad (1)$$

where IP refers to the instruction prompt and R_i denotes as the rationale of (Q_i, P_i, A_i) .

Sub-graph matching. After obtaining the rationales R , we aim to compute the similarity score $S_i = \{S_{i1}, S_{i2}, \dots, S_{in}\}$ between each rationale $R_i \in R$ and all n entities in MMKG. Specifically, given a rationale R_i and entities $E = \{e_1, e_2, \dots, e_n\}$, we first employ a pre-trained text knowledge retriever to embed R_i and entities’ textual descriptions $T = \{T_1, T_2, \dots, T_n\}$. Then, we calculate the cosine similarity between the embeddings of R_i and every entity description $T_j \in T$. Finally, for each entity $e_j \in E$, we add its cosine score with the maximum cosine

score of e_j 's neighboring entities to obtain S_{ij} , which represents the final similarity score between R_i and e_j . This process is formulated as:

$$C_i = \{C_{ij} | C_{ij} = \cos(T - KR(R_i), T - KR(T_j)), T_j \in T\}, \quad (2)$$

and

$$S_{ij} = C_{ij} + \alpha * \max(\{C_{ik} | C_{ik} \in C_i, e_k \in E_j\}), \quad (3)$$

where C_{ij} serves as the e_j 's cosine score to R_i , $T - KR$ denotes the pre-trained text knowledge retriever, E_j means the set of e_j 's neighboring entities, and α is a pre-defined weight hyperparameter. We perform the above process on all questions in the training set and obtain the question-knowledge similarity score set $S = \{S_1, S_2, \dots, S_m\}$, where m represents the number of questions in the training set. S serves as the supervision signal for subsequent training of the multi-modal knowledge retriever.

4.2. Multi-modal knowledge retriever training

After obtaining the question-knowledge similarity score set, we aim to train a multi-modal knowledge retriever in this section. To this end, we introduce a medical multi-modal model as the retriever and train it with two kinds of loss functions. The first contrast learning loss function helps the retriever better distinguish medical images of different entities in the MMKG. The second KL divergence loss function requires the retriever to output similarity scores that have a similar distribution to the obtained similarity score set for the given question and entity knowledge. The detailed descriptions of these loss functions are as follows.

Contrastive loss computing. The contrastive loss L_{cl} is built as a $K+1$ -way classification problem [41]. Given an image I_j of the entity e_j as the query, we first sample K images of other entities as the negative samples and another image of e_j as the positive sample. Then we employ the multi-modal knowledge retriever to embed these images. Finally, we compute the contrastive loss based on cosine similarity between the image embeddings. This process can be formulated as the follows:

$$\text{sim}(I_j, I^{+/-}) = \cos(M - KR(I_j), M - KR(I^{+/-})), \quad (4)$$

and

$$L_{cl} = -\log \left(\frac{e^{\text{sim}(I_j, I^+)/\tau}}{e^{\text{sim}(I_j, I^+)/\tau} + \sum_{k=1}^K e^{\text{sim}(I_j, I_k^-)/\tau}} \right), \quad (5)$$

where I^+ and I^- represents the positive and negative sample respectively, $M - KR$ denotes as the multi-modal knowledge retriever, and, τ is a temperature parameter.

KL divergence loss computing. Given an MMKG and a triple of the question, medical image and corresponding similarity score (Q_i, P_i, S_i) , we first employ the multi-modal knowledge retriever (M-KR) to embed all entities' textual description T , entities' images I , Q_i and P_i . Then, for every entity e_j in the MMKG, we respectively multiple the embedding of its textual description T_j with Q_i , and the embedding of its image I_j with P_i . The sum of the two multiplication results is served as the similarity score S'_{ij} between e_j and the given question. Finally, we perform the above process to all n entities in the MMKG and obtain the similarity score set $S'_i = \{S'_{i1}, S'_{i2}, \dots, S'_{in}\}$, which can be mathematically represented as:

$$S'_i = \{S'_{ij} | S'_{ij} = M - KR(Q_i) * M - KR(T_j) + M - KR(P_i) * M - KR(I_j), j \in (1, n)\}. \quad (6)$$

After obtaining S'_i , we first consider S'_i and S_i into vector form and normalize them using Softmax. Then we calculate the KL loss based on these two similarity distribution vectors. This process is formulated as follows:

$$L_{kl} = f_{KL}(f_{softmax}(S'_i), f_{softmax}(S_i)), \quad (7)$$

where $f_{KL}(\cdot)$ is a function that returns the KL divergence between the distributions of two given vectors. We train the multi-modal knowledge retriever by minimizing the above two loss functions.

Table 1

The statistic of the constructed MMKG.

Statistic	Organ	Disease	Overall
Entities	106	303	409
Images	7592	12,953	20,545
Triples	384	2522	2906
Relation categories	13	48	61
Description length (Avg)	81	95	91

4.3. Retrieval augmented generation

After obtaining a trained multi-modal knowledge retriever, we employ it and an MMKG to enhance the LVLM's performance in the MedVQA task. The core idea is that given a question and a medical image, we first employ the knowledge retriever to obtain the related entity description from the MMKG. Then we feed the question, medical image and retrieved descriptions into the LVLM to achieve the final answer.

Specifically, given an MMKG, a question Q_i and a medical image P_i from the test set D_{test} , we first employ the multi-modal knowledge retriever to embed Q_i , P_i and the entities' textual description T and images I in MMKG. Then, we follow Eq. (6) to compute the similarity score between the question and entities. Finally, we select the textual descriptions of entities with top- p similarity scores and concatenate them with Q_i and P_i as the input of LVLM. The output of the LVLM is the final answer A_i . This process can be formulated as follows:

$$T'_i = \{T_j | T_j \in T, S_{ij} \in \text{Top}_p(S_i), S_i = f_{sqe}(Q_i, P_i, T, I), Q_i \in D_{test}\}, \quad (8)$$

and

$$A_i = LVLM(Q_i, P_i, T'_i), \quad (9)$$

where $f_{sqe}(\cdot)$ denotes as Eq. (6), returning the similarity scores between entities and the given question. S_{ij} represents the entity e_j 's similarity score and T_j is e_j 's textual description.

5. Multi-modal medical knowledge graph

In this section, we first introduce the construction process of the MMKG. Then, we present its detailed statistics and analysis of image quality and computational cost.

5.1. MMKG construction

We construct the MMKG based on a public medical knowledge graph (MKG) [22], which has 52.6K triples of the head entity, relation and tail entity. The construction process consists of three main steps as follows.

Textual description generation. In this step, we aim to obtain the textual description of the entities in the MKG. Given an entity of the organ or disease (e.g., Pertussis), we first extract its attribute triples (e.g., <Pertussis, symptom, spastic cough>) from the MKG. Then we employ GPT-4¹ to generate the entity textual descriptions based on the collected multiple attribute triples. The used instruction prompt is that "Please write a brief summary for the following entity and its attributes."

Image collection. In this step, we aim to obtain the images of the entities in the MKG. Given an entity of the organ or disease, we regard it as the query and collect the corresponding images from three sources: a web search engine (Google images) and two public medical image datasets ROCO² and PubMed³.

¹ <https://chat.openai.com/>.

² <https://huggingface.co/datasets/MedIR/roco>.

³ <https://huggingface.co/datasets/FreedomIntelligence/PubMedVision>.

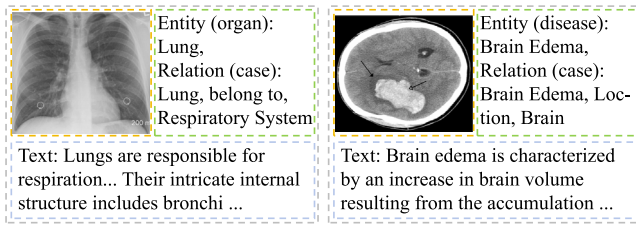


Fig. 3. The entities and their corresponding images, relations, textual descriptions in the constructed MMKG.

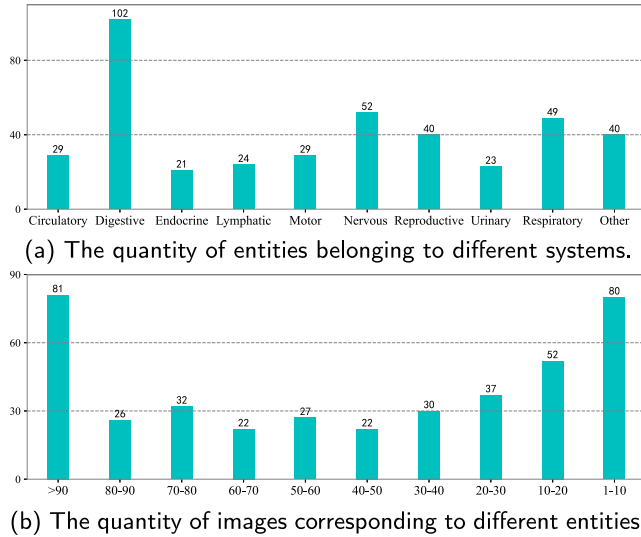


Fig. 4. The distributions of medical entities and collected images in the constructed MMKG.

Image filtering. Since there may exist noisy images that not depict the corresponding entity, we introduce a pre-trained medical multi-modal model BiomedCLIP [8] to remove these images. Given an entity e_j , we first employ BiomedCLIP to embed its textual description T_j and images. Then, for each image I_j , we compute the cosine similarity between the embedding of T_i and I_j . This process can be formulated as:

$$\text{sim}(T_j, I_j) = \cos(\text{B-CLIP}_{\text{text}}(T_j), \text{B-CLIP}_{\text{visual}}(I_j)), \quad (10)$$

where B-CLIP denotes as BiomedCLIP that consists of a separate textual encoder and a visual encoder. Finally, we remove images with a similarity score below the pre-defined threshold of 0.28.

5.2. MMKG analysis

We first present the statistic of the constructed MMKG. Then we evaluate the quality of the collected images and analyze the computational cost.

Statistic. Table 1 presents the statistics of entities, images, and triples in the constructed MMKG. As shown in Table 1, our MMKG consists of 106 organ entities and 303 disease entities, with 2906 triples used to build the relations between them. For the entities, each has a corresponding textual description and 50 images on average (see Fig. 3). For the triples, there are 61 relation categories to establish associations between organ and disease entities. To have a better understanding of the constructed MMKG, we show the distributions of the medical entities and collected images in Fig. 4. It is observed that most medical entities in MMKG are belong to the digestive system and have at least 10 corresponding images.

Image quality analysis. To verify the quality of the collected images in MMKG, we employ manual and automatic evaluation. For manual evaluation, we invite three domain experts. Following the previous work [42], the evaluation criteria is that if an image reflects the corresponding entity, it is labeled as 1. Otherwise, it is labeled as 0. We randomly select 1800 images of 409 entities for evaluation. To ensure the evaluation quality, we additionally include 200 noisy images that do not correspond to the given entity and check whether the evaluators successfully find them. Finally, the three evaluators respectively recall 1.0, 0.97 and 0.98 of these noisy images and the weighted average accuracy is 0.89. The Fleiss's kappa [43] reaches to 0.81, showing the consistency of the manual evaluation. Hence, the collected images in MMKG are of high quality. For automatic evaluation, we apply the collected images to the MedVQA task to demonstrate their effectiveness. The details are reported in Section 6.4.

MMKG computational cost. In this paper, the primary time consumption for constructing the MMKG lies in collecting candidate images for medical entities and filtering them with the BiomedCLIP model. During the image collection process, we spend approximately 0.74 h to achieve 40,478 candidate images of entities. In the image filtering process, with a single RTX4090 GPU, we spend around 0.37 h performing text-image matching calculations on these images and remain 20,545 of them. On average, it only takes about 0.2 s obtain one entity image. Therefore, constructing the MMKG is of low time cost. For maintaining the MMKG, it only requires reusing the previous processes to collect and filter images for new medical entities. Therefore, the cost of maintaining the MMKG also remains manageable.

6. Experiment

In this section, we first report the detail of our experimental setup and the results of MKGF compared to baseline methods. Then we provide the ablation study and detailed analysis to demonstrate the importance of each component.

6.1. Experimental setup

Datasets. To evaluate the effectiveness of the proposed MKGF framework, we conduct experiments on two public MedVQA datasets: (1) Slake [22] and (2) VQA-RAD [44]. For Slake, we use its English version, which contains 642 radiology images and 7033 question-answer pairs. For VQA-RAD, it contains 315 radiology images and 3515 question-answer pairs.

Baselines. We compare the proposed MKGF with recent non-RAG methods and other state-of-the-art (SOTA) knowledge-based RAG methods. The baselines include:

1. **Med-Flamingo** [18]: A non-RAG method that introduces relative question-answer pairs for LVLMS to perform few-shot learning based inference.
2. **CoMT** [19]: A non-RAG method that first prompts LVLMS to hierarchically analyze the medical visual question and then answer it step by step.
3. **Wikichat** [45]: A method that uses an LVLMS to judge whether knowledge retrieval is needed and to specify what knowledge is required, followed by retrieving the knowledge based on the LVLMS's output.
4. **KAT** [33]: A method that first employs a multi-modal model to compute the similarity between question image region features and entities, and then retrieves entity knowledge with high similarity score.
5. **Wiki-Llava** [20]: A method that first uses a multi-modal model to retrieve entity descriptions similar to the question image, then selects text chunks highly relevant to the question text from them.

Table 2

The performance of LVLMs with different RAG methods on the slake and VQA-RAD dataset. “Base” means the result of LVLMs answer the question without any external knowledge. “Silver knowledge” approximates the theoretical best performance of combining the given LVLm and MMKG, which simulates the question–entity similarity computing step and employs the rationale of the question–answer pair to retrieve entity knowledge.

Model	Method	Slake			VQA-RAD		
		Close	Open	Overall	Close	Open	Overall
Llava -Med-7B	Base	62.02	43.26	52.64	67.73	36.88	52.31
	Med-Flamingo	62.26	44.23	52.99	68.13	39.01	53.57
	CoMT	63.38	43.71	53.55	69.72	38.41	54.06
	WikiChat	65.63	41.43	53.53	72.11	38.36	55.24
	KAT	67.79	44.09	55.94	69.32	39.73	54.53
	Wiki-LLava	67.07	45.62	56.35	69.32	39.57	54.45
	ViQuAE ^a	69.47	44.55	57.01	73.71	42.34	58.02
	CRKVQA ^a	70.91	45.46	58.19	73.31	43.36	58.34
	MKGf(Ours) ^a	71.88	52.42	62.15	75.70	45.40	60.55
	Silver knowledge	75.72	55.49	65.60	77.69	47.36	62.53
HuatuoGPT -Vision-7B	Base	76.20	51.97	64.09	73.71	51.04	62.37
	Med-Flamingo	77.16	52.94	65.05	74.50	52.63	63.57
	CoMT	79.29	55.46	67.38	74.10	53.33	63.72
	WikiChat	76.68	59.34	68.01	72.32	58.06	65.19
	KAT	77.16	59.79	68.48	76.10	57.71	66.90
	Wiki-LLava	77.64	60.35	69.00	76.89	58.88	67.89
	ViQuAE ^a	80.05	63.03	71.54	79.68	57.08	68.38
	CRKVQA ^a	79.33	63.39	71.36	80.88	56.74	68.81
	MKGf(Ours) ^a	82.69	65.83	74.26	81.67	61.72	71.69
	Silver knowledge	82.93	69.66	76.30	83.67	61.51	72.59

^a Methods indicates that entity images from MMKG are used as visual cues when retrieving entity knowledge.

6. **ViQuAE** [37]: A method that first employs multiple mono-modal retrievers to compute the visual or textual similarity scores between the question and entities, then retrieves knowledge based on these scores.

7. **CRKVQA** [38]: A method that combines the mono-modal and cross-modal similarity scores between the question and entities to retrieve the knowledge.

Evaluation Metrics. We evaluate the proposed MKGF following the setting of the previous work [2]. Specifically, for close-ended questions, we evaluate the performance by measuring the accuracy of ground-truth tokens present in the generated responses. In the case of open-set questions, we use recall to assess the proportion of ground-truth tokens that are included in the generated answers. Since we do not impose constraints on responses to open-set questions, our approach aligns more closely with the open-set nature of the task, but it inherently presents greater challenges.

Implementation details. We introduce two pre-trained medical LVLMs for our research: LLava-med-7B⁴ and HuatuoGPT-Vision-7B⁵. HuatuoGPT-Vision is a SOTA pre-trained LVLm for MedVQA. During question–entity similarity computing, we choose BGE⁶ as the pre-trained knowledge retriever and set α as 0.2 for sub-graph matching. During multi-modal knowledge retriever training, we choose Biomed-CLIP as the multi-modal knowledge retriever and optimize it using AdamW [46] with a learning rate of $1e-4$. We set the K as 4, τ as 0.5, batch size as 4, and the training epoch is 10. During the process of retrieval augmented generation, we retrieve the descriptions of top-3 high-similarity entities as the external knowledge for LVLms. To ensure the stability of the results, we set the temperature to 0.0 during inference with LVLm.

6.2. Main result

In this section, we compare the performance of MKGF with all baseline methods on the Slake and VQA-RAD datasets. The results are detailed in Table 2.

Based on the experimental results, we can draw the following observations and conclusions. (1) All the RAG methods achieve a better performance than the based method on both datasets. MKGF improves the performance of HuatuoGPT-Vision-7B by 10.15% and 9.32% without finetuning its abundant parameters. This shows that incorporating external knowledge from the MMKG enhances the capability of the LVLm on the MedVQA task. (2) Although recent non-RAG methods can further improve the ability of LVLms to perform MedVQA, there remains a significant performance gap compared to MKGF. The primary reason is that MKGF can supplement LVLms with the missing medical entity knowledge, while these methods cannot. (3) Overall, methods that use the MMKG’s images as clues during the retrieval process perform better than those that do not. Specifically, MMKG (HuatuoGPT-vision) achieves an improvement of over 5.26% and 3.80% on the two datasets compared to Wiki-LLava. This demonstrates the effectiveness of our collected entity images in MMKG. (4) Our proposed method MKGF achieves better performance than all the baseline RAG method with different LVLms on both datasets. The main reasons are that we design a contrastive loss to help the multi-modal retriever distinguish images of different entities, and address the challenge of medical images corresponding to multiple entities with a KL Divergence loss and a sub-graph matching algorithm. (5) All methods still show a performance gap compared to silver knowledge. Hence, how to train a better retriever to provide more relevant entity knowledge for the LVLm is a future research direction.

6.3. Ablation studies

In this section, we present an ablation analysis of our method to assess the impact of each component. The MKGF’s variants include: (1) w/o L_{cl} : multi-modal retriever training without the contrastive loss function, (2) w/o L_{kl} : multi-modal retriever training without the KL divergence loss function, (3) w/o SBGM: question–entity similarity computing without sub-graph matching and (4) w/o Visual: retrieval augmented generation without the visual clue.

From Table 3, we conclude that: (1) Removing any component results in a performance decline, demonstrating their importance. (2) Both two loss function play a role in enhancing the performance of multi-modal knowledge retriever. Compared to MKGF, w/o L_{cl} results in a decrease of over 2.24% and w/o L_{kl} leads to a reduction of

⁴ <https://github.com/microsoft/LLaVA-Med>.

⁵ <https://github.com/FreedomIntelligence/HuatuoGPT-Vision>.

⁶ <https://huggingface.co/BAAI/bge-large-en-v1.5>.

Table 3
Ablation studies of MKGF.

Dataset	Slake		
Methods	Open	Close	Avg
MKGF(HuatuoGPT)	82.69	65.83	74.26
w/o L_{cl}	80.05	63.38	71.71
w/o L_{kl}	78.13	58.21	68.17
w/o SBGM	78.85	62.14	70.49
w/o Visual	79.57	55.81	67.69

Dataset	VQA-RAD		
Methods	Open	Close	Avg
MKGF(HuatuoGPT)	81.67	61.72	71.69
w/o L_{cl}	80.88	58.03	69.45
w/o L_{kl}	77.69	54.03	65.86
w/o SBGM	80.48	57.75	69.11
w/o Visual	76.10	52.14	64.12

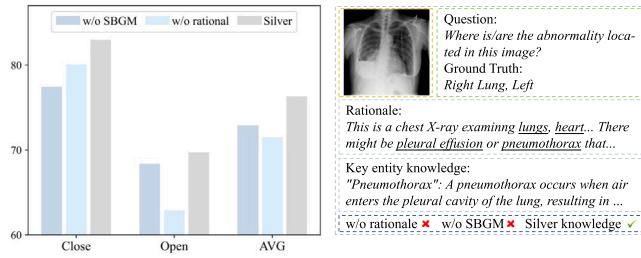


Fig. 5. The performance of silver knowledge (HuatuoGPT-Vision-7B) with or without the rationale and sub-graph matching based retrieving on Slake.

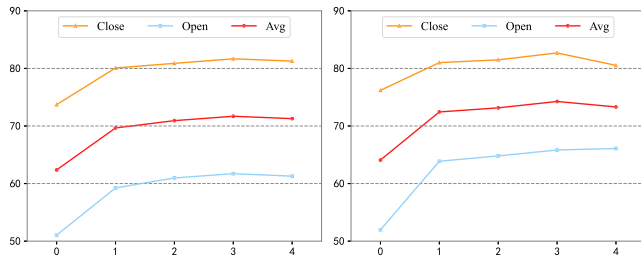


Fig. 6. MKGF (HuatuoGPT-Vision-7B) with different top-p settings on VQA-RAD (left) and Slake (right).

more than 5.83% in the average score across two datasets, respectively. Hence, we can infer that L_{cl} contributes to distinguishing images of different entities and L_{kl} helps build the connection between question and entities. (3) Without the sub-graph matching in question–entity similarity computing, MKGF’s performance drops on all metrics. Hence, sub-graph matching algorithm helps obtain better training data for the multi-modal retriever, addressing the challenge of medical images corresponding to multiple entities. (4) Compared to MKGF, w/o Visual’s performance significantly declines on both datasets. This demonstrates that the visual information of entity images in MMKG and medical question images helps the multi-modal retriever obtain the entity knowledge relevant to the given question.

6.4. Detailed analysis

In this section, we present a detailed analysis of the components in the MKGF framework, knowledge sparsity, fine-tuned LVLMS in MedVQA, and the collected entity images in MMKG.

Analysis of question–entity similarity. To evaluate the effectiveness of rationale generation and sub-graph matching in question–entity similarity, we compared the performance of silver knowledge with and without these steps on Slake. As shown in Fig. 5 (left), the performance

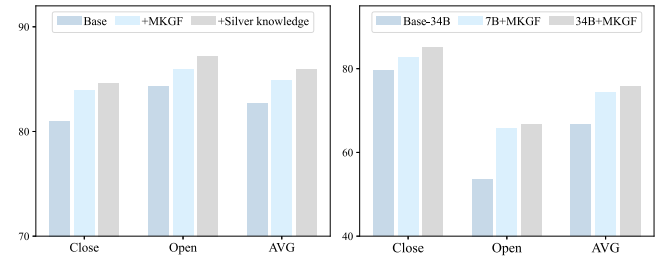


Fig. 7. The performance of MKGF with Llava-med-7B-fine-tuned (left) and HuatuoGPT-Vision-34B (right) on Slake.

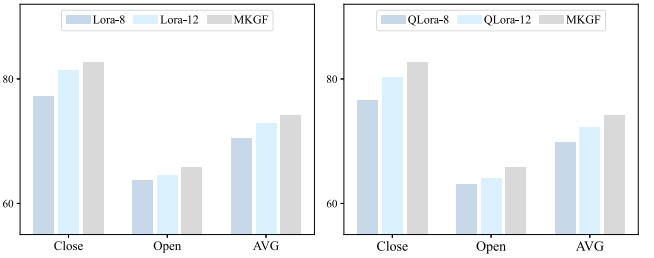


Fig. 8. The performance of MKGF(HuatuoGPT-Vision-7b) compared to the fine-tuned version of HuatuoGPT-Vision-7b using Lora (left) and QLora (right) on Slake.

Table 4

The performance of ViQuAE and CRKVQA (HuatuoGPT-Vision-7B) with different kinds of retrieval clues on Slake.

Methods	Close	Open	Avg
ViQuAE + text	78.13	55.08	66.60
ViQuAE + noise	71.39	56.34	63.87
ViQuAE + our	80.05	63.03	71.54
CRKVQA + text	76.68	54.26	65.47
CRKVQA + noise	74.76	46.03	60.39
CRKVQA + our	79.33	63.39	71.36

drops more than 3.43% after removing either rationale generation or sub-graph matching. Hence, we can conclude that these two steps contributes to retrieving question-relevant entity knowledge, thereby providing higher-quality data for the subsequent multi-modal retriever training. To intuitively understand the effectiveness of these steps, we show a case in Fig. 5 (right). It is observed that the key entity knowledge *pneumothorax* are unable to be retrieved with BGE after removing these steps.

Analysis of different Top-p entity knowledge setting in retrieval augment generation. To assess the impact of the choice of parameter p in the retrieval augment generation step, we record the performance of MKGF under different settings of p . From Fig. 6, we have two observations. (1) The performance of the MKGF framework improves progressively as p increases, showing that the retrieved entity knowledge starts to relieve the LVLMS hallucination issue. This also demonstrates that the medical visual questions often involves the knowledge of multiple entities. (2) The performance of MKGF tends to converge when p grows to 4. Therefore, we set p as 3, balancing performance and time cost. To better understand how the retrieved Top-p entity knowledge helps LVLMS relieve hallucination issues, we show some cases in Fig. 9.

Analysis of MKGF with different kinds of LVLMS. To evaluate the robustness of MKGF with different kinds of LVLMS, we introduce two models: Llava-med-7B-finetuned and HuatuoGPT-Vision-34B. The experimental results presented in Fig. 7 show that: (1) MKGF enables to enhance the performance of either fine-tuned or larger size LVLMS, and (2) MKGF combining a smaller LVLMS with 7B parameters achieves

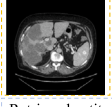
 Question: Where is/are the abnormality located? Ground Truth: Liver. Retrieved entity knowledge: "Liver": The liver is a organ situated in the upper right quadrant of the abdomen... "Liver Tumors": Liver tumors are neoplastic lesions found in the liver... LLava-med: The abnormality is located in the right adrenal gland... LLava-med+MMKG: The abnormality is located in the liver, specifically in right ...	 Question: What is the main organ in the image? Ground Truth: Lung. Retrieved entity knowledge: "Lung": Lungs are a part of the chest and are vital organs. The function of lung is to... "Chest": The chest is the area of the body located between the neck and abdomen... LLava-med: The main organ is the heart, specifically the left ventricle... LLava-med+MMKG: The main organ in the image is the lung ...	 Question: Do the organs in the image exist in the head? Ground Truth: Yes. Retrieved entity knowledge: "Cerebellum": The cerebellum is a crucial brain structure located in intracranial ... "Inner Ear": The inner ear comprises two main structures: the bony labyrinth... HuatuoGPT: No. This is an axial CT scan of the abdomen and pelvis region showing... HuatuoGPT+MMKG: Yes, the organs in the image exist within the head...	 Question: Which part of the body does this image belong to? Ground Truth: Abdomen. Retrieved entity knowledge: "Liver": The liver is a organ situated in the upper right quadrant of the abdomen ... "Gallbladder": The gallbladder is a small, pear-shaped organ behind the liver... HuatuoGPT: This image belongs to the thoracic region, showing heart and lungs... HuatuoGPT+MMKG: This image belongs to the abdomen region of the human ...
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Fig. 9. Cases of how retrieval augmentation benefits LVLMs on MedVQA. It is observed that LVLm (LLava-med-7B or HuatuoGPT-Vision-7B) generates a wrong answer without the retrieved entity knowledge while LVLm+MMKG generates a correct answer after retrieval augmentation.

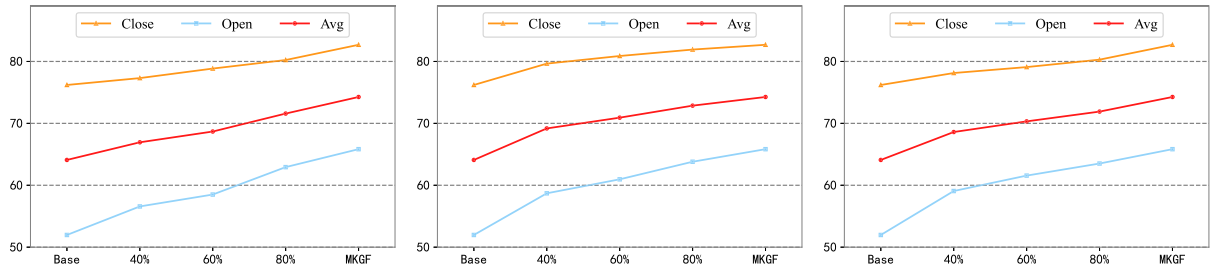


Fig. 10. Analysis of data sparsity. The performance of MKGF (HuatuoGPT-Vision-7b) with an incomplete MMKG on slake, considering the remaining of {40%, 60%, 80%} entities (left), relations (middle) and images (right).

a better performance than LVLm's 34B version. These findings demonstrate that MKGF contributes to mitigating the hallucination issues of various LVLms on the MedVQA task.

Analysis of data sparsity. To evaluate how knowledge sparsity within the MMKG impacts the MKGF's performance, we first randomly delete entities, relations or image information in the constructed MMKG and then compare the performance of MKGF with different incomplete MMKGs ({40%, 60%, 80%} completeness). From Fig. 10, we can observe two conclusions. (1) The greater the completeness of the knowledge in the MMKG, the better the MKGF framework performs. Specifically, randomly deleting 20% of the MMKG entities, relations, or images resulted in an impact of about 2% on the performance of MKGF. (2) MKGF can enhance the performance of LVLm by more than 3% with an MMKG that has only 40% completeness.

Analysis of MKGF compared to fine-tuned LVLms. To evaluate the performance difference between MKGF and fine-tuned LVLms, we introduce two high-efficient fine-tuning methods Lora [47] and QLoRA [48]. The performance of the fine-tuned HuatuoGPT-Vision-7B under different rank settings {8 and 12} is shown in Fig. 8. It is observed that MKGF with a pre-trained LVLm achieves an improvement of over 1.3% compared to its fine-tuned version. This demonstrates the effectiveness of MKGF.

Analysis of image quality in MMKG. For automatic evaluation of image quality in MMKG, we design an A/B testing following the previous work [42]. The used information of the multi-modal knowledge retriever for experiment A is the entity descriptions (+text) while experiment B has two kinds of information: (1) entity descriptions and entity images in MMKG (+our) and (2) entity descriptions and entity images with disrupted correspondences (+noise). We compare the performance of two knowledge-based RAG frameworks, ViQuAE and CRKVQ, with different types of retrieved information on the Slake dataset. As shown in Table 4, both knowledge-based RAG frameworks with (+our) outperform those using only textual information, indicating the importance of image features for entity knowledge retrieval. Moreover, both methods with (+Noise) achieve no significant improvement, demonstrating that the improvement is mainly due to the collected entity images rather than the added visual encoder. Hence, we conclude that the collected entity images in MMKG are effective.

7. Conclusion

In this paper, we construct a multi-modal medical knowledge graph and propose a novel framework named MKGF to combine it with the LVLm for MedVQA. This framework first introduces an LVLm and a pre-trained retriever to obtain question–entity matching relationships on the training set. Then, it employs these relationships to train a multi-modal retriever with a contrastive loss and a KL divergence loss. Finally, the trained multi-modal retriever is used to obtain the relevant entity knowledge from the MMKG and help the LVLm answer the medical visual question on the test set. Extensive experiments are conducted on two public MedVQA datasets to demonstrate the effectiveness of MKGF.

In our experiments, we observe that the potential limitations of applying MMKG to the MedVQA task are mainly reflected in two aspects. Firstly, this method requires the MMKG to have a high level of completeness and quality. The performance of the MKGF framework drops after removing parts of the entities, relations or images in the MMKG. Hence, how to construct a high-quality MMKG is a direction for future works. Secondly, this method requires on the capabilities of the LVLm itself. The MKGF framework with HuatuoGPT-Vision achieves a better performance than with LLava-med. The better the LVLm, the more effectively it can understand and utilize the knowledge from MMKG. Therefore, how to pre-train a better LVLm is a meaningful direction for future research.

CRedit authorship contribution statement

Yinan Wu: Writing – original draft, Methodology, Investigation. **Yuming Lu:** Validation, Software. **Yan Zhou:** Software. **Yifan Ding:** Validation. **Jingping Liu:** Writing – review & editing, Conceptualization. **Tong Ruan:** Supervision, Project administration.

Ethical statement

Since some images in the constructed MMKG are sourced from internet, we need to clarify two main ethical statements: (1) image

safety and (2) image copyright. For image safety, we believe that our MMKG does not include harmful images (e.g., violent images). This is because we employ the Google SafeSearch content filter to remove the harmful images and the queried medical entities are reliable (sourced from a public medical knowledge graph). For image copyright, we use Google Image URLs to filter out images without copyright license agreement. Our MMKG is released under Creative Commons Non-Commercial (CC-NC), prohibiting commercial use.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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